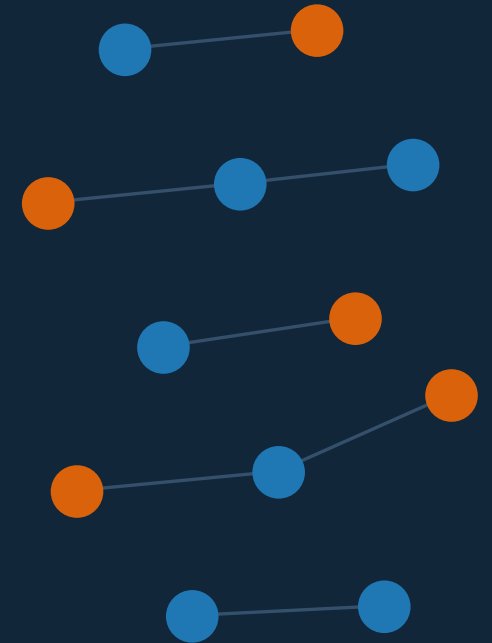


Protein Language Models Are Accidental Taxonomists

Hallee, Peleg, Rafailidis & Gleghorn · BMC Bioinformatics (2026) 27:188

Burak Suyunu

LifeLU Reading Club · 18.09.2026

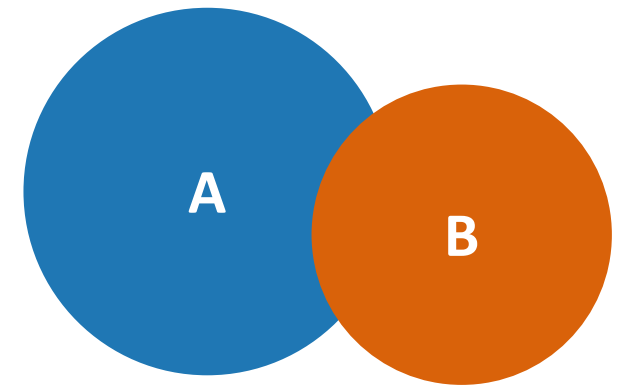


What a protein–protein interaction is

- Two proteins in physical contact, mediating a chemical or conformational change.
- At scale these form the interactome — the network underlying signalling, regulation, and most disease mechanisms.
- Measured by Y2H, co-IP/MS, cell-free two-hybrid. Slow, expensive, each with its own bias profile.

**But a PPI is a statement about a context,
not a property of two molecules.**

PTMs · chaperones · cofactors · pH · localisation · the assay itself



observed once, under some conditions,
by some method

What the prediction task actually is

Input: two amino-acid sequences

Output: one binary label

Scale

~20,000 human proteins → >200 million possible pairs. A vanishingly small fraction interact. Experiment cannot cover this space.

Metrics

Binary classification: accuracy, F1, MCC, ROC AUC, PR AUC. MCC is the one that degrades honestly under imbalance and threshold shifts.

The benchmark question

Does this model generalise to protein pairs it has never seen? Splits are the battleground — and the field has partly fixed the obvious leakage.

Hold that shape in your head: two sequences in, one logit out.

Two numbers

~0.65

accuracy · ~0.30 MCC

best published models,
rigorous single-species benchmarks

>0.90

accuracy · 0.8–0.9+ AUC

same architectures,
multi-species datasets

Same task. Same model families. Same era.

The case

More species → more data → more diversity → better models.

Reasonable. But a 25-point jump? The authors' word is “staggering.”

The accidental taxonomist hypothesis

Multi-species PPI models inflate their scores by detecting whether two proteins come from the same species — not whether they interact.

There are no negatives

**You cannot experimentally prove that two proteins never interact.
Non-interaction is not an observable.**

- Negatome 2.0 — literature mining, manual annotation, structural analysis.
- Against hundreds of thousands of curated positives.
- Train on the raw data and any binary classifier collapses to predicting “positive.”

~6,000

negative pairs
across all Negatome splits combined

The negative class has to be invented. Every problem in this paper is downstream of how.

So negatives get invented. Two ways.

Localisation-based

Pair proteins from different subcellular compartments.

Assumption: they never meet in vivo.

Random pairing

Pair proteins with no recorded interaction. (current default)

Assumption: arbitrary sequences don't bind.

Prior conviction — “accidental localizers” (same group, SYNERACT)

pLM embeddings predict subcellular localisation accurately → a model trained on localisation-derived negatives can learn localisation, not interaction.

If your negatives are defined by property X, and embeddings encode X, your model can score well by learning X.

The field's response: Bennett's gold standard

Human-only. Positives from HIPPIE, negatives by random pairing.

1

No node-degree bias

PPI networks are scale-free. Hubs leak “is this protein promiscuous” as a proxy for interaction.

2

No sequence overlap between splits

C3 (Park & Marcotte): no protein crosses splits, not just no pair.

3

Max 40% sequence identity

Enforced across train / validation / test.

Note for later: the authors adopt C3 and 40% identity for their own splits — but not degree control.

The honest ceiling

On the gold standard, across many architectures and several years:

~0.65

accuracy

~0.70

AUC

~0.30

MCC

“Deep learning models for unbiased sequence-based PPI prediction plateau at an accuracy of 0.65.”

Reim et al., Bioinformatics 2025 — the paper title

Controlled single-species: 0.30 MCC. Uncontrolled multi-species: >0.90 accuracy.
Both claim to predict the same thing. The difference is the dataset.

Shuffle the columns

- BioGRID 4.4.243 — positives only, IDs mapped to UniProt, organism recorded per protein.
- Independently shuffle the protein-A and protein-B columns. $\times 100$.
- **This is exactly how the field generates negatives.**

Positives

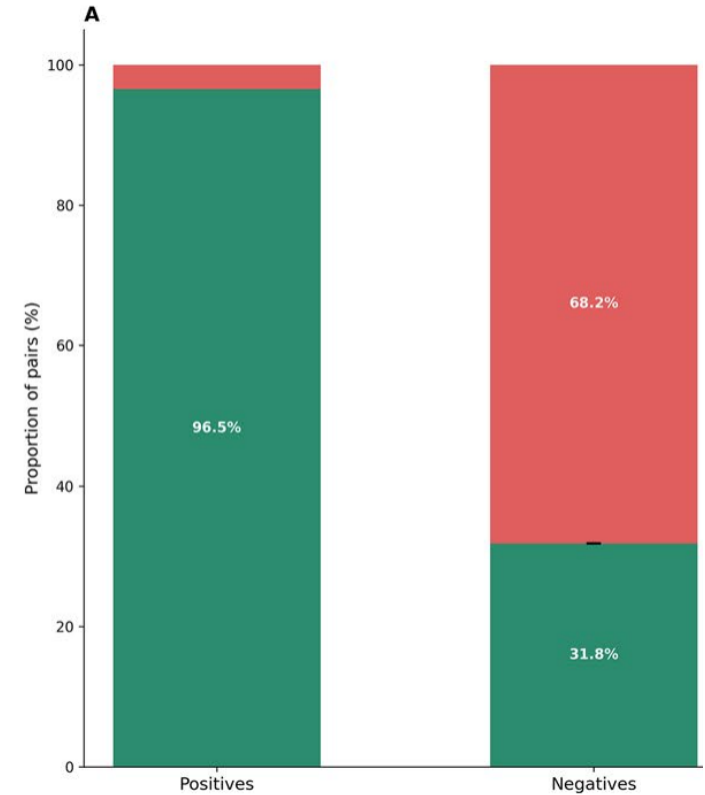
96.5%

intra-species

Shuffled negatives

31.8%

$\pm 0.02\%$ over 100 shuffles

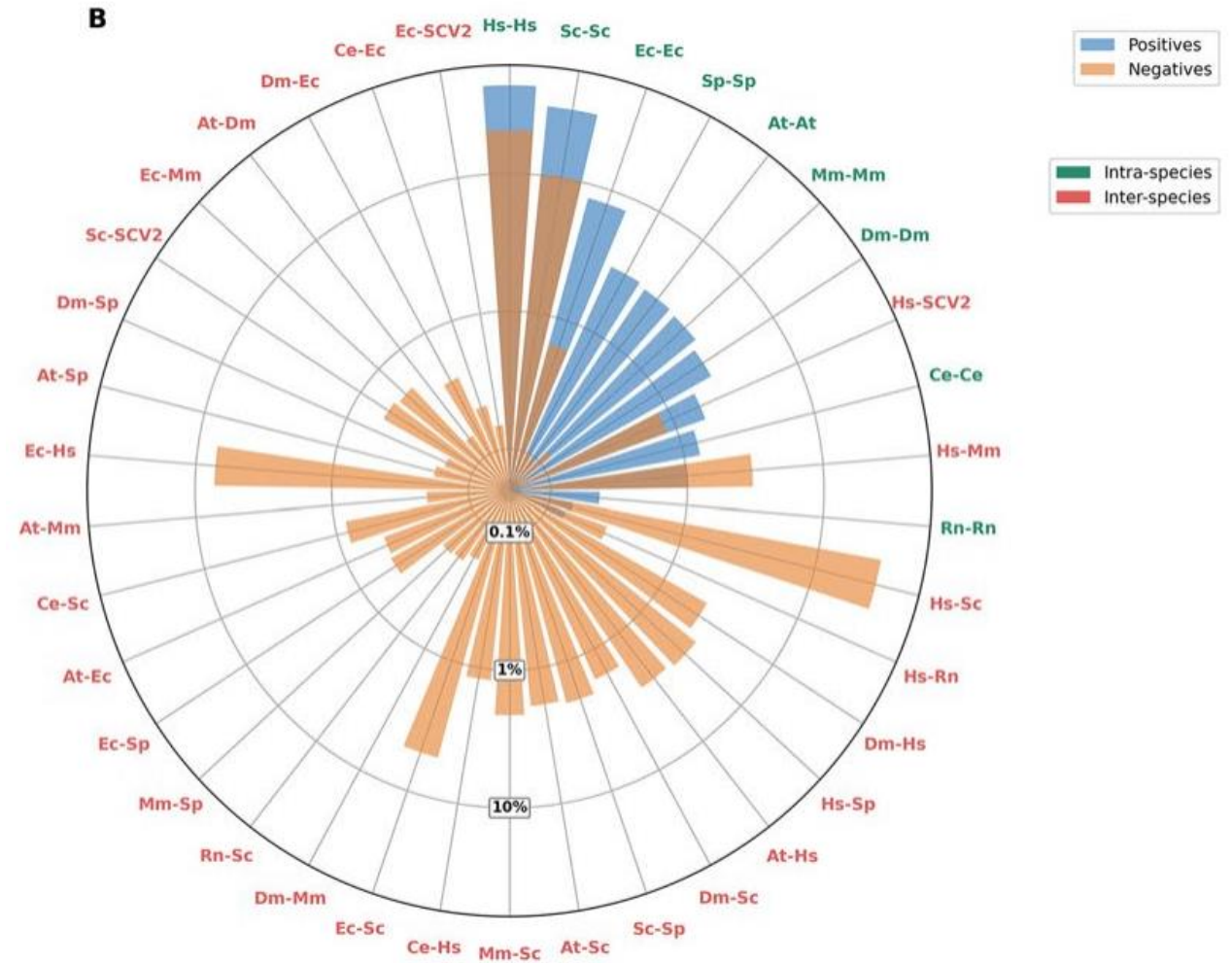


Hallee et al. 2026, Fig. 1A

The same picture, per organism pair

- Positives concentrate on the diagonal: Hs–Hs, Sc–Sc, Ec–Ec
- Negatives fan onto the off-diagonal: Hs–Sc, Ec–Hs
- BioGRID is >44% human, remainder dominated by a few model organisms.

The label correlates with the species pair,
not with the proteins.



A classifier that knows nothing about binding

Suppose a model learns only: same organism, yes or no? Predict “interacts” $\Leftrightarrow$ same organism.

Catches 96.5% of positives. Correctly rejects 68.2% of negatives. On a balanced 100,000-example set:

0.824

accuracy

0.845

F1

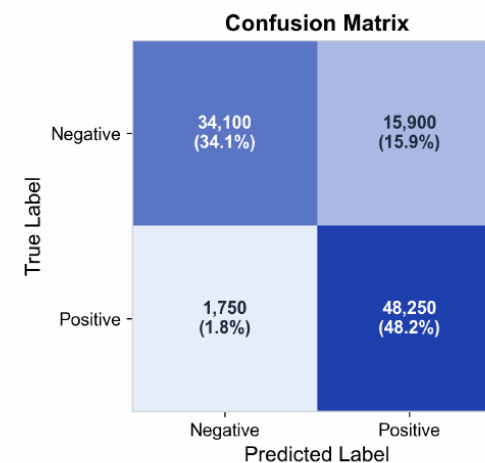
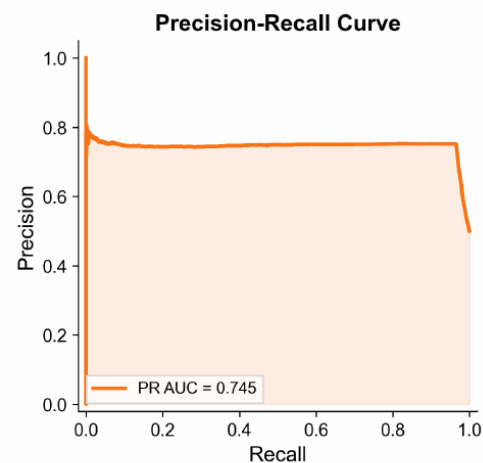
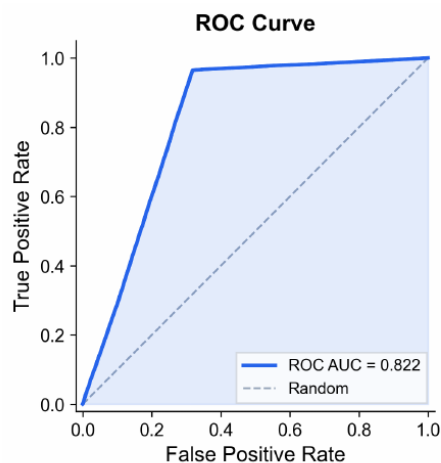
0.822

AUROC

0.675

MCC

precision 0.752 · recall 0.965 · specificity 0.682 · PR AUC 0.745



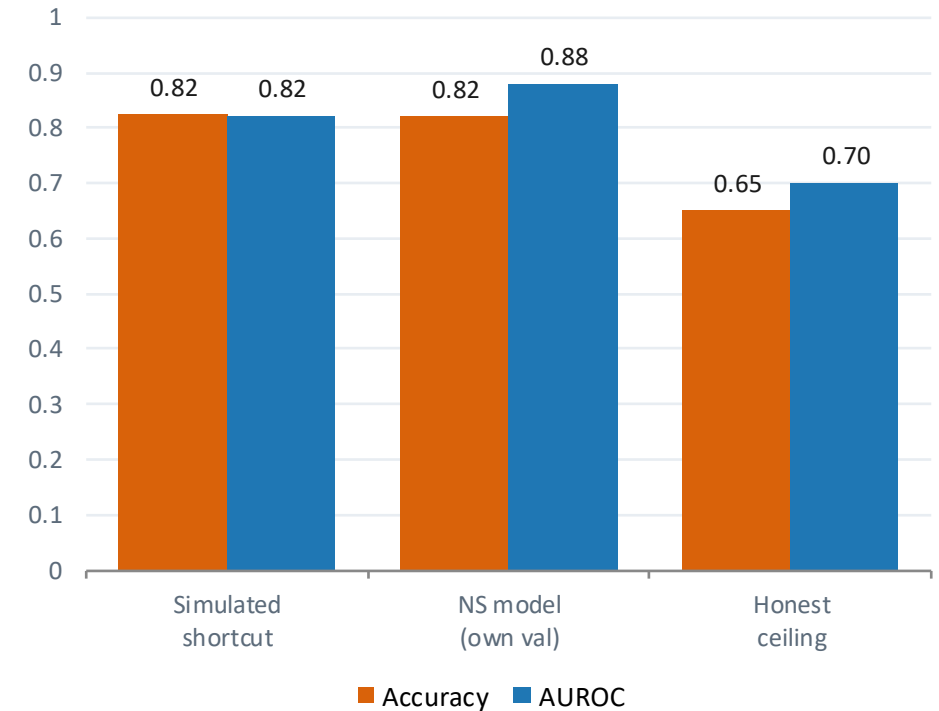
Against the literature

	Accuracy	AUROC
Pure taxonomic shortcut (simulated)	0.824	0.822
NS model, its own validation set	~0.82	~0.88
Honest single-species ceiling	~0.65	~0.70

Not a comparison the paper makes. NS values read off Supp. Fig. S2 — approximate.

The simulated pure-taxonomy classifier and the actual trained NS model land in nearly the same place on the same metrics.

Not proof of identity — but it is why the hypothesis deserves the next twenty minutes.



What the simulation does and doesn't show

**The simulation shows the shortcut is available.
It does not show that any model takes it.**

A Methods detail

- A perfect taxonomic classifier is deterministic — it would give a three-point ROC curve.
- The AUROC and PR AUC come from injected uniform noise: 0.51–1.00 for true positives, 0.00–0.49 for true negatives, 0.51–1.00 for false negatives.
- **Accuracy, precision, recall, F1 and MCC are unaffected.**

Everything so far is a claim about the dataset, not about any model's behaviour. The paper closes that gap next.

Can a pLM actually see species?

Task A

single protein → taxonomic label

multi-class, domain ... species

Task B

protein pair → same species?

binary

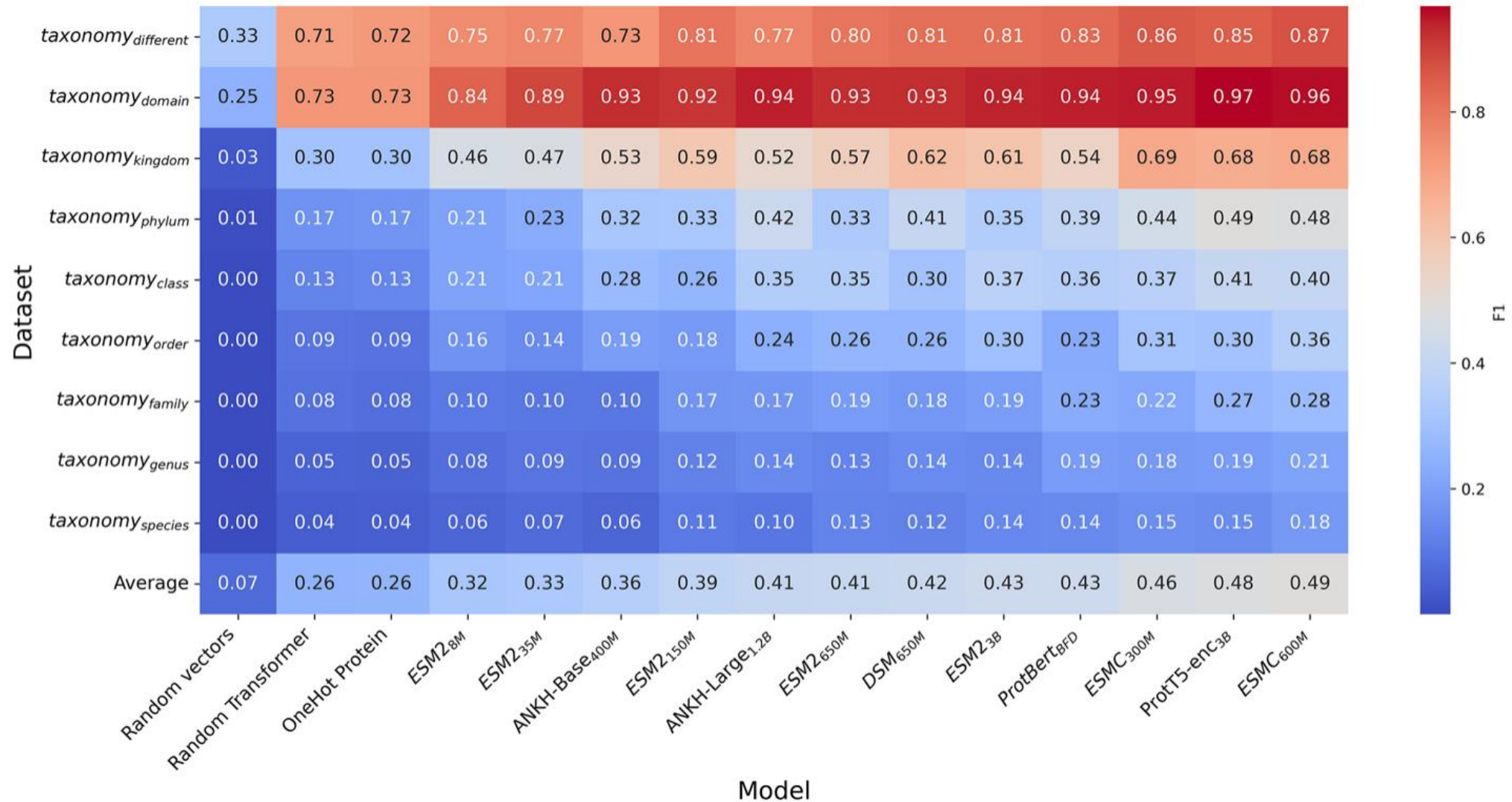
Dataset and probe

- Swiss-Prot 2025_03 · lengths 20–2048 · deduplicated · CD-HIT 40% (word size 2)
- Clusters assigned to splits; no sequence or cluster overlap
- Labels kept only where ≥ 100 examples: 3 domain → 618 species classes
- Mean + variance pooled last hidden states → frozen-feature MLP probe (h = 8,192)

The probing results

Hallee et al. 2026, Fig. 2 · weighted F1

domain 0.97 → kingdom 0.69 → phylum 0.49 → class 0.41 → order 0.36 → family 0.28 → genus 0.21 → species 0.18 (random: 0.00)



The inversion

Which species is this protein from?

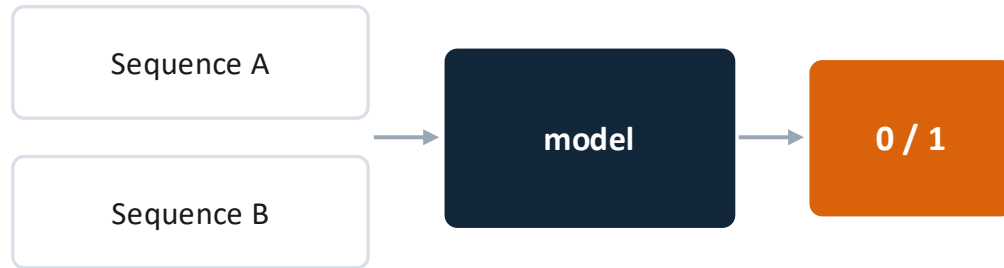
0.18 F1

Are these two proteins from the same species?

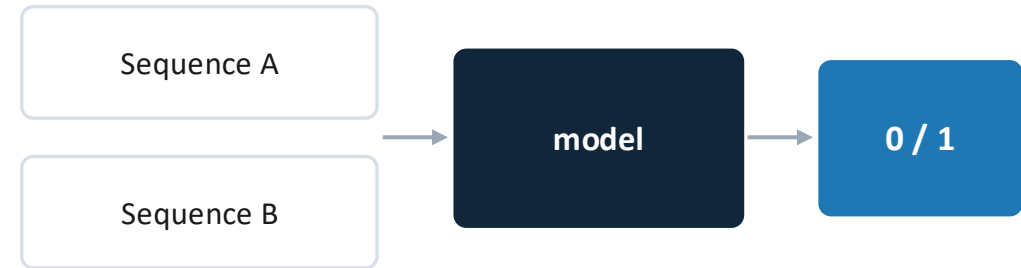
0.87 F1

Why that specific task

taxonomy_different



PPI prediction



Identical tensor shapes. Identical loss. Nothing in the architecture distinguishes them.

The difference lives entirely in the labels — and in BioGRID those labels are 96.5% / 31.8% correlated with species identity.

The experiment nobody ran

Established

pLM embeddings contain species information.

Established

NS models behave as if they use it.

Never tested: does the PPI model's internal representation encode species?

Not done

- Probe the penultimate layer for species identity
- Correlate the output logit with phylogenetic distance between the input pair
- Ablate

All three are cheap on models they already trained.

The finding the paper buries

On taxonomy_different:

0.87

ESMC-600M

~0.72

one-hot encoding

~0.71

random-init transformer

A one-hot encoder with no pretraining, no semantics and no evolutionary knowledge gets most of the way to a frontier pLM.

If the confound is largely amino-acid composition, it isn't a pLM problem.

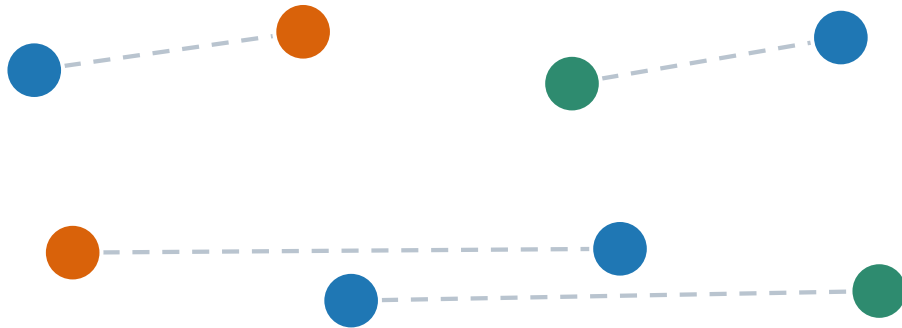
k-mer features · classical ML · CNNs · a decade of pre-pLM PPI literature — all had access to this shortcut.

The paper is more damaging to the field than its title admits.

Two ways to sample a negative

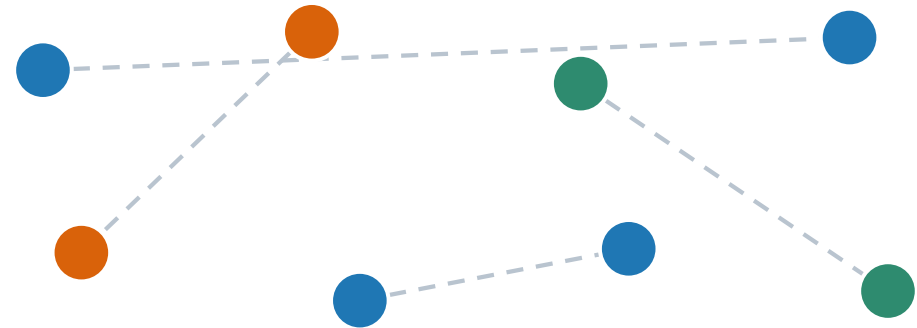
NS — Normal Sampling

random pairs from the whole multi-species dataset



SS — Strategic Sampling

random pairs from within the same species



One change. Every pair in the dataset — positive or negative — is now intra-species.

The taxonomic signal is still in the embeddings. It is no longer predictive.

Experimental design

4,523,432

training examples

10,070

validation

10,034

test

76,045

unique sequences

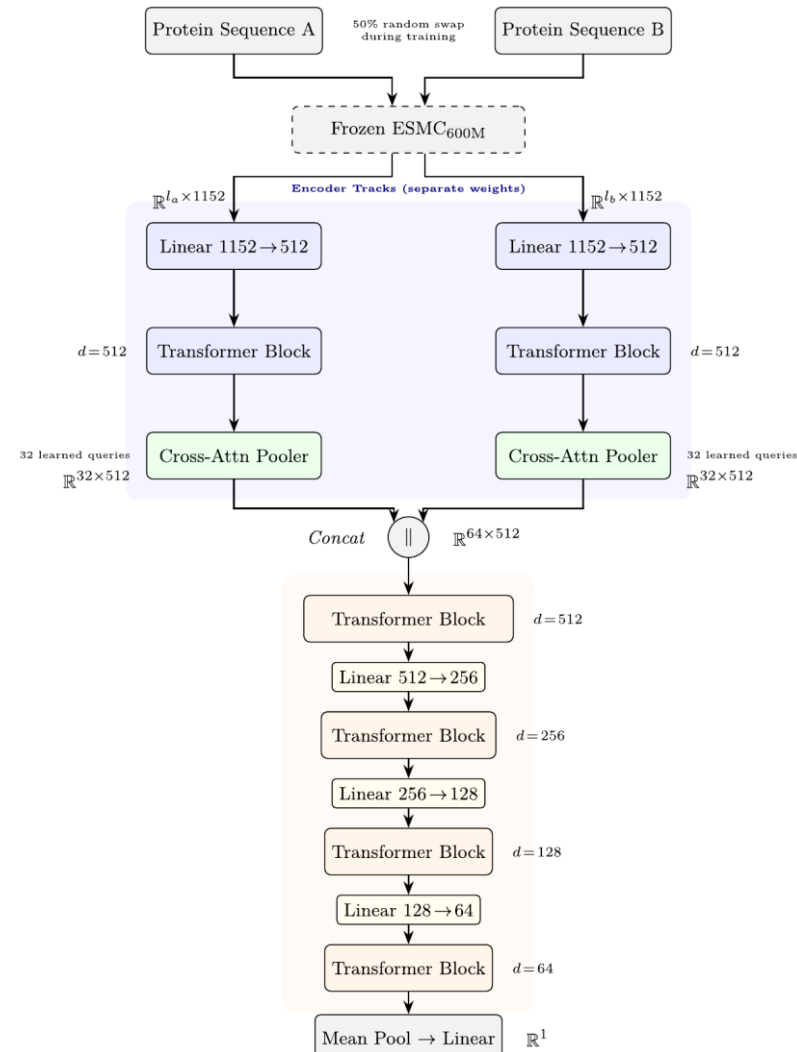
- Identical dataset sizes in both conditions · perfectly balanced labels · C3 splits · CD-HIT 40%
- Candidate negatives cross-referenced against all known positives, discarded and resampled if found

Both models are evaluated on the same SS test set.

Each validates on its own distribution. Both are tested where the shortcut is unavailable.

It is a trap: train on NS, test where the shortcut can't operate, see what's left.

The model



- Frozen ESMC-600M, sequences truncated to 512 residues
- Two non-Siamese encoder tracks → transformer block (SwiGLU) → cross-attention pooler → 32 tokens each
- Concatenate to 64×512 → 4 transformer blocks, $512 \rightarrow 256 \rightarrow 128 \rightarrow 64$ → mean pool → linear head
- 50% random A/B orientation swap · BCE · AdamW · batch 128 · lr $1e-4$ · 1 epoch · single GH200
- Threshold chosen by scanning 100 points on the PR curve for max F1

The authors' framing: the accidental taxonomist is a data issue, not a model issue.

They expect the same NS/SS gap with any reasonable architecture.

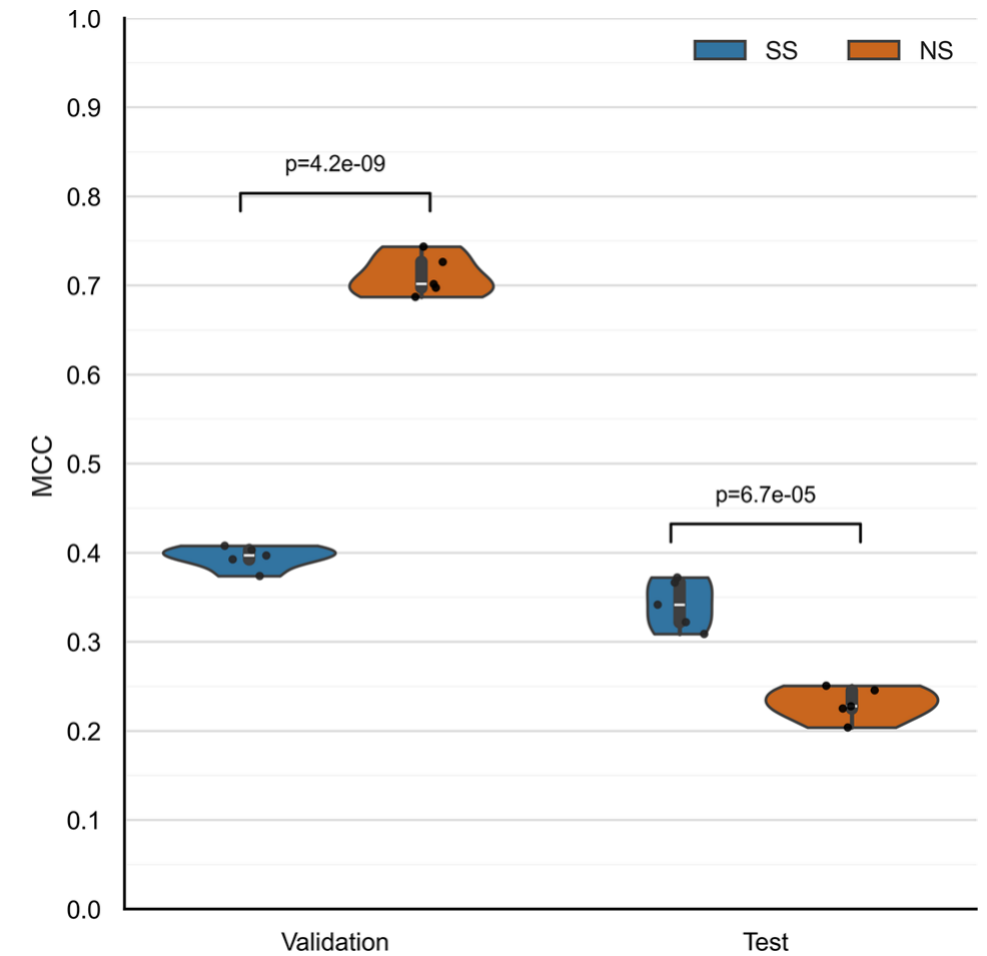
Non-Siamese tracks give each protein its own subspace (they argue this matters for homodimers); concatenation preserves independent information downstream.

The collapse

MCC	own validation	shared SS test
NS	0.71	0.23
SS	0.39	0.34

t-test $p < 0.0001$ for both differences, $n = 5$ seeds

On validation, NS looks nearly twice as good.
On the shared test set, the ordering reverses.



Hallee et al. 2026, Fig. 3

What's left when the shortcut is gone

62%

accuracy

0.61

F1

0.63

AUROC

NS at 0.23 MCC on the SS test set — still clearly above chance.

taxonomic shortcut

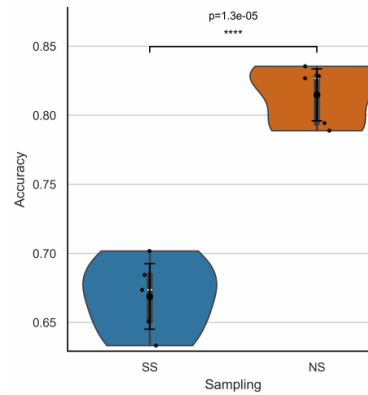
genuine PPI signal

Illustrative only — the decomposition was never measured.

NS models do learn some real PPI signal. It was swamped, not absent.

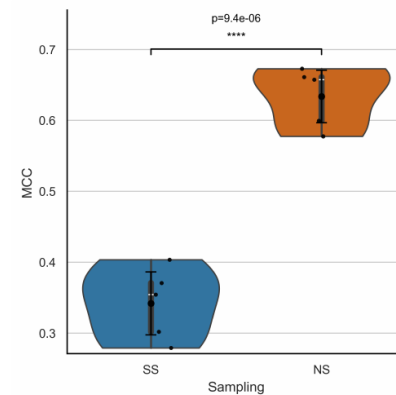
Every metric, same reversal

Validation • Accuracy



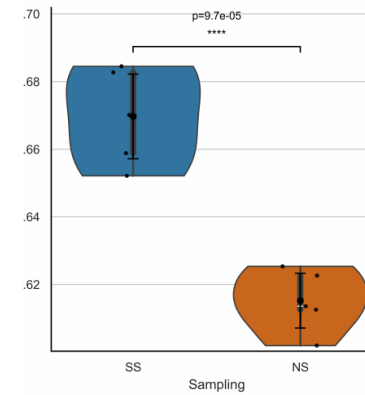
(a) Accuracy

Validation • MCC



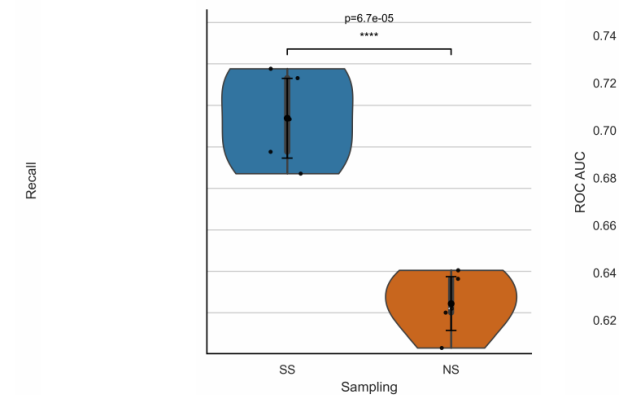
(e) MCC

Test (SS) • Accuracy



(a) Accuracy

Test (SS) • MCC



(e) MCC

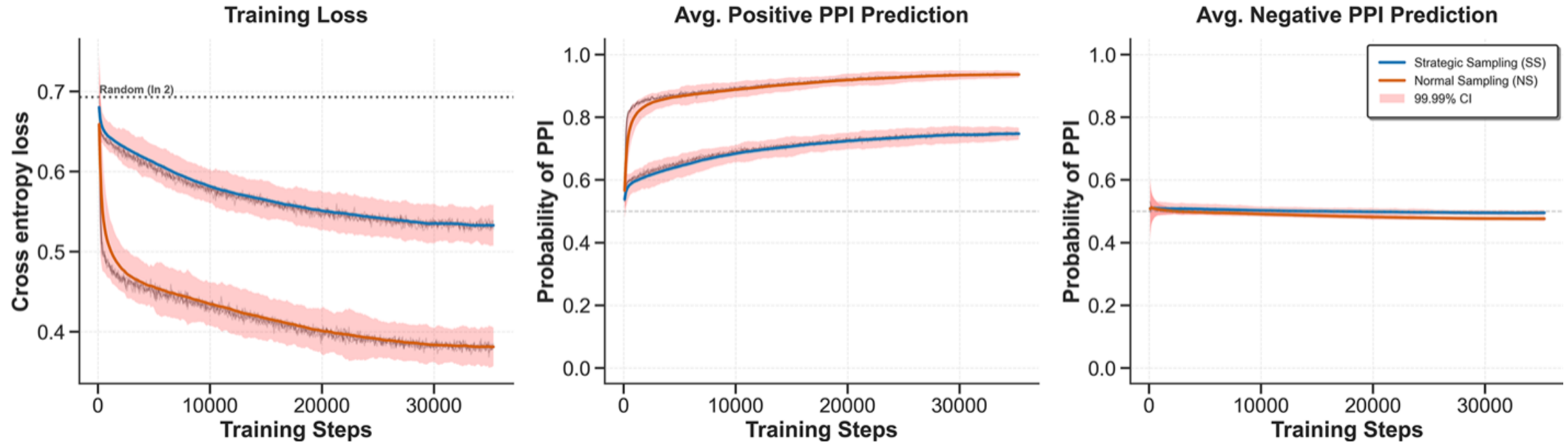
Hallee et al. 2026, Supplementary Figs. S2 (validation) and S3 (test)

Validation: NS wins accuracy, recall, precision, F1, MCC, ROC AUC, PR AUC — p from $1e-5$ to $1e-7$.

Test on shared SS: every single one of them flips.

This is not a metric artefact or a threshold artefact.

Training dynamics



Hallee et al. 2026, Fig. 4

Positive examples

NS → 0.94 SS → 0.75

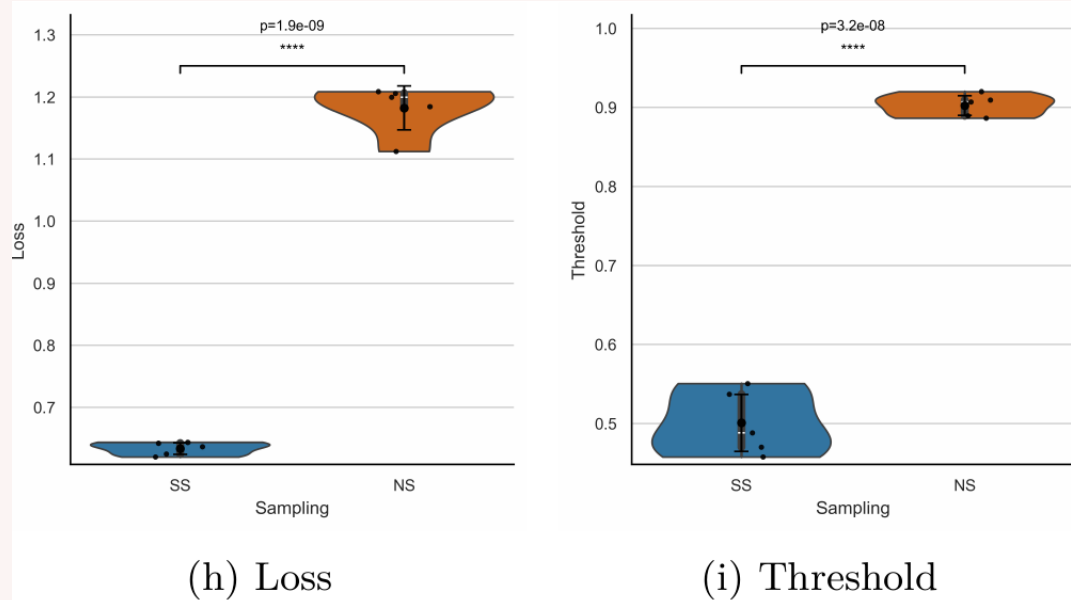
Negative examples

both ≈ 0.50

Loss trajectories

completely disjoint from very early in training

The detail buried in the supplement



Hallee et al. 2026, Supp. Fig. S3 (h), (i)

NS cross-entropy on the SS test set

≈ 1.2

random baseline ($\ln 2$)
0.693

Optimal decision threshold

SS 0.5

NS 0.9

Removing the shortcut doesn't make the NS model mediocre.

It makes it confidently wrong.

SS is not a clean control

SS negatives are two random proteins from the same organism — far more likely to be real but uncatalogued interactions.

So the NS → SS drop confounds two changes: less shortcut, and dirtier labels.

The authors' defence (they raise this themselves)

Magnitude

0.71 → 0.23 is too large for label noise alone.

Asymmetry

If noise drove it, SS models should suffer too. Instead they are stable: 0.39 → 0.34.

Persuasive — but it is an argument, not a measurement. Inject matched label noise into NS and see how much of the gap you recover.

Two more things I'd push on

What controls their own splits?

C3 + 40% CD-HIT ✓ node-degree control: not reported.

PPI networks are scale-free. Is 0.34 MCC the honest number — or the next inflated one?

The weakest headline claim

“Multi-species SS models exceed the 0.3 MCC single-species plateau.” SS best 0.37 vs Bennett 0.30

Different database · different test set · ~45% positive overlap · vastly larger training set

Every caveat is in the Discussion, hedged as “suggestive, not definitive.”

It's the Abstract that states it cleanly — and abstracts are what get cited.

What about interactions that really do cross species?

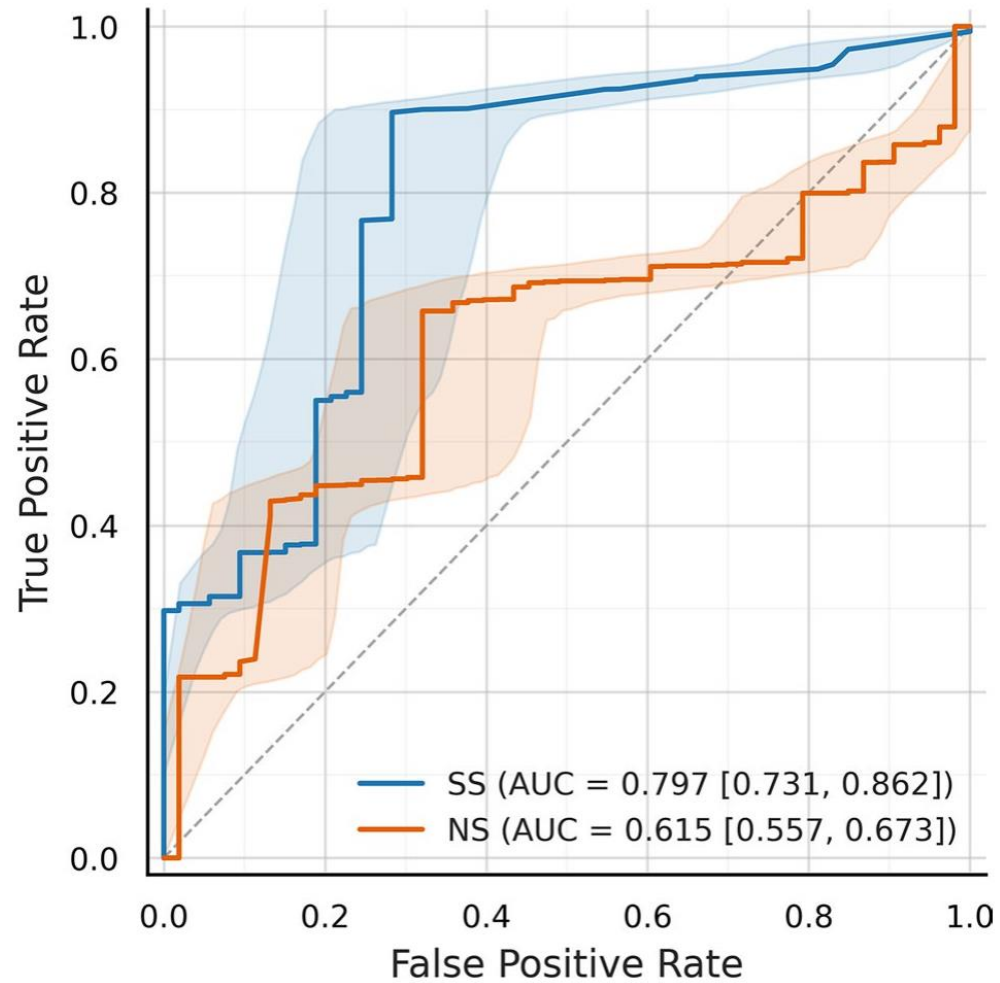
Viral entry mechanisms. Bacterial effectors. Host–pathogen biology.

These are inter-species positives — and the whole argument so far has treated inter-species as a proxy for negative.

BioGRID Multi-Validated v5.0.253

- Every example validated by ≥ 2 methods from ≥ 2 sources
- 35,970 inter-species positives — about 7% of 531k pairs
- Inter-species-aware C3 split, MMseqs2 at 40% identity / 80% coverage
- New NS and SS models, shared SS test set

Results on BioGRID-MV



Hallee et al. 2026, Fig. 5

Test set: 2,815 examples — 2,762 positives, 53 negatives

SS AUROC **0.80** [0.73 – 0.86]

NS AUROC **0.62** [0.56 – 0.67]

DeLong $Z = 3.60$, $p = 3.2 \times 10^{-4}$

Hold on to that negative count.

The actual result

Inter-species positives in the test set

1,091

Identified by the SS model

0

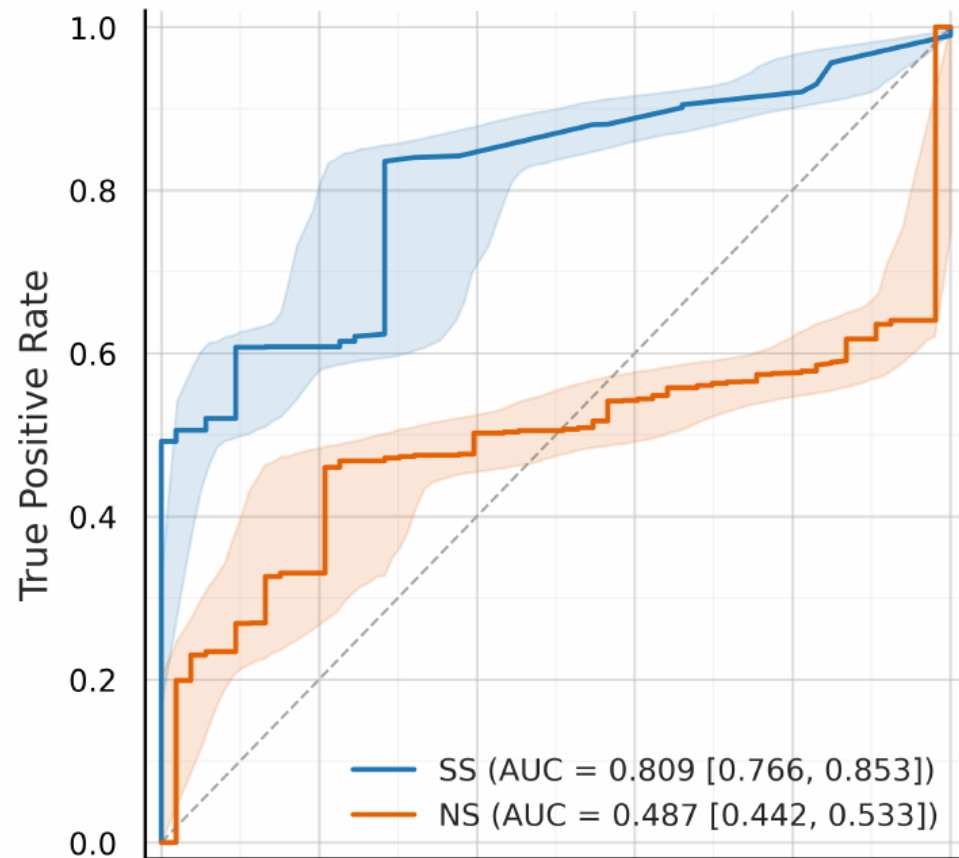
Identified by the NS model

2

Mean predicted probability, both models: 0.46

And the shortcut story holds underneath

Restricted to the intra-species subset (n = 1,724; 1,671 positives, 53 negatives)



Hallee et al. 2026, Supp. Fig. S4

SS

0.809

[0.766 – 0.853]

NS

0.487

[0.442 – 0.533]

The NS confidence interval contains 0.5.
Not degraded — indistinguishable from guessing.

DeLong Z = 11.37, $p < 10^{-15}$

Scope — and what survives it

53 negatives

in both experiments — hence the wide bands; the SS interval nearly touches the diagonal in places.

982 of 1,091

inter-species positives are human–SARS-CoV-2.

Supported: models trained this way fail on SARS-CoV-2–human pairs.

Overreach: inter-species PPI is invisible.

But the 0 / 1,091 result doesn't touch the negatives at all — it is pure recall. The AUROC comparison is fragile. The blindness finding is not.

Even the model that cannot cheat fails here. Fixing negative sampling does not fix host–pathogen PPI.

What the paper establishes

Demonstrated

- The taxonomic bias exists and is large — 96.5% vs 31.8%
- pLM embeddings encode same-vs-different species at 0.87 F1
- NS models drop to ~chance when the shortcut is removed
- Both sampling regimes are blind to inter-species positives

Suggested, not demonstrated

- Multi-species beats single-species when properly controlled
- That models are mechanistically using taxonomy
- That SS performance is itself confound-free (mine, not the paper's)

This is not a PPI problem

Enzyme Commission numbers

If an EC class occurs mostly within one taxonomic family, the model can classify by taxonomy instead of function.

Applies to any supervised protein task — including regression, where a label range may map onto a clade.

Our labels reflect what researchers chose to study.

Study effort is wildly uneven across the tree of life.

→ The correlation between labels and phylogeny is closer to the default than the exception.

Nobody is safe

Simple models too

A cross-attention model over two proteins is the obvious risk.

But a feed-forward net can dedicate a weight layer to per-class average taxonomic features. Gradient descent will happily store training-distribution structure in weights.

Single-input architectures are not immune.

Genomic LMs should be worse

Codon usage bias predicts taxonomic origin on its own.

DNA and codon-vocabulary models stack that channel on top of the amino-acid-level features already present in pLM embeddings.

The authors' speculation, not a result.

Both points follow from the same fact: the signal is in the data, not in the architecture.

“Accidental _____”

accidental localizer

negatives defined by subcellular compartment

accidental taxonomist

negatives defined by species

accidental ?

Any unintended physical, chemical, biological — or social — regularity in how proteins got selected for study.

What to actually do

- 1 Report the taxonomic composition of positives and negatives.
- 2 Sample negatives within whatever class defines your positives.
- 3 Build a test slice where the confound is absent by construction, and report it beside your headline number.
- 4 Include composition controls — one-hot, shuffled labels, random-init transformer.
- 5 **Treat >0.95 AUC / >0.8 MCC / >0.9 F1 as a red flag, not a result.**

Items 1–4 are lifted from their methodology; item 5 is the authors' own stated heuristic.

**The model was right.
The reason was wrong.**

Nobody noticed, because the metric didn't care.